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Bypassing ISP: Transfer Learning-Based Object Detection on Raw Bayer Images with Faster R-CNN

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***Abstract*—Object detection models traditionally rely on RGB images, which are processed through computationally intensive image signal processing (ISP) pipelines to enhance visual quality. However, such pipelines can reduce the fidelity of raw image details, potentially affecting detection accuracy. This paper presents a novel transfer learning-based framework to enhance object detection performance directly on raw Bayer images. Our approach leverages a pre-trained Faster R-CNN model trained on RGB images, followed by selective fine-tuning of the last five layers of the ResNet-50 backbone while keeping the earlier layers frozen to adapt the model to the unique characteristics of Bayer images. Experimental results on the KITTI dataset and its generated Bayer dataset demonstrate that our method significantly outperforms traditional Faster R-CNN on Bayer images and achieves performance comparable to RGB-based models. This framework provides an effective solution for low-power edge computing by bypassing ISP and directly processing raw Bayer images without increasing model complexity.**

***Index Terms*—Object Detection, Transfer Learning, Faster R-CNN, Raw Image**

I. Introduction

In the field of computer vision, image signal processing (ISP) has long been a critical step in converting raw sensor data into interpretable images [1]. Traditional advanced vision tasks, such as object detection, segmentation, and recognition, typically rely on RGB images generated through ISP pipelines [2]. These pipelines are designed to enhance the visual quality of images by performing operations such as demosaicing, noise reduction, color correction, and gamma adjustment, making them easier for humans to interpret and for deep learning models to process [3]. This workflow has been highly effective in numerous applications, laying the foundation for state-of-the-art performance in various computer vision benchmarks.

Although ISP is traditionally viewed as essential, some research [4], [5] suggests that directly applying raw Bayer images to high-level tasks can also be feasible and beneficial [6]. Several studies have demonstrated that bypassing ISP retains more low-level information, such as subtle color and texture details, that might otherwise be lost or distorted during ISP transformations [7], [8]. Leveraging raw sensor data can therefore unlock new potential for precision in advanced vision tasks, particularly in scenarios where the accuracy of fine-grained details is critical. However, the challenges associated with processing raw Bayer data include the need for task-specific adaptations and the complexity of handling unprocessed data formats [5].

Meanwhile, transfer learning has become a cornerstone of modern deep learning, offering significant advantages in scenarios where labeled data is scarce or tasks are computationally intensive [9]. By utilizing pre-trained models developed for related tasks, transfer learning enables rapid adaptation to new objectives, significantly reducing training time and improving model performance [10]. This approach has proven especially effective in resource-constrained environments, making it an ideal candidate for addressing the challenges of raw Bayer image processing.

Motivated by these advancements, we propose a novel framework that integrates the strengths of ISP-based methods and transfer learning to enhance object detection on raw Bayer images. Specifically, we aim to address the limitations of traditional RGB image pipelines while leveraging the transferability of learned features. Our approach first trains a Faster R-CNN [11] model on conventional RGB images to establish a robust feature extraction backbone. Subsequently, we freeze most parameters in the backbone before the Feature Pyramid Network (FPN) [12] and fine-tune the subsequent layers and output heads on raw Bayer data. This two-step strategy bridges the gap between ISP-processed and raw image domains, enabling effective utilization of raw sensor data for object detection while capitalizing on the efficiencies of transfer learning.

In summary, this work makes the following key contributions:

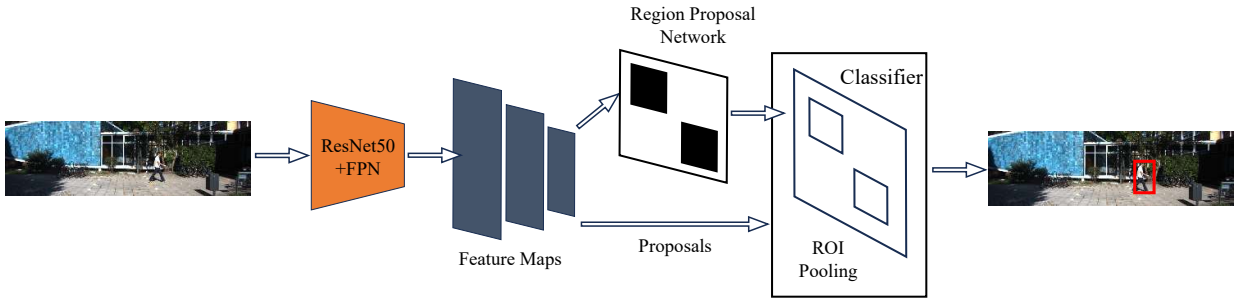
- We propose a transfer learning framework designed for object detection on raw Bayer image, effectively enhancing the performance on unprocessed sensor data.
- We systematically explore the value and potential of transfer learning in object detection tasks involving raw Bayer data, providing significant insights and references for future research in this area.

II. Related Work

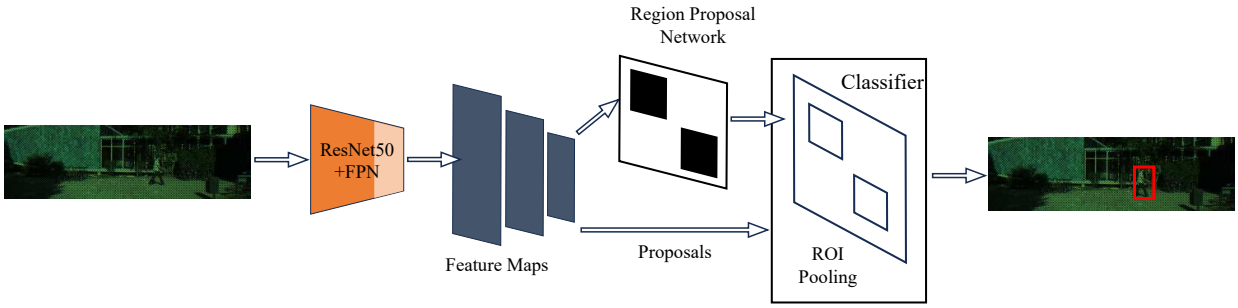
A. ISP-Free Approaches for Object Detection on Raw Images

Recent advancements have explored bypassing traditional Image Signal Processing (ISP) pipelines specifically in object

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(a) Traditional Faster R-CNN



(b) Modified Faster R-CNN with Transfer Learning and Fine-Tuned Heads

Fig. 1. Structure of Faster R-CNN and Modified Faster R-CNN

detection tasks. [13] demonstrated that object detection models can maintain competitive performance when trained on raw images without ISP processing, highlighting the potential computational and energy savings of such approaches. Complementing this, [2] observed that images with less visual appeal, which are closer to raw sensor data, may provide better results for object detection tasks compared to traditionally processed, visually pleasing images. Recognizing the limitations of traditional ISP, [14] proposed an evolutionary algorithm-based framework that optimizes ISP components to cater specifically to high-level vision tasks, achieving both cost-effectiveness and improved accuracy. [15] further challenged the conventional step-by-step ISP pipeline by introducing an end-to-end CNN for image processing, which yielded improved image quality tailored to computer vision applications.

Specific applications of ISP-free object detection include vehicle detection and tracking on Bayer images. For instance, [16] evaluated YOLOv5 and DeepSort on raw Bayer images, laying the groundwork for future ISP-free advancements. Meanwhile, [5] extended Faster R-CNN with additional camera parameters and specialized fully connected layers, achieving performance gains.

Beyond eliminating ISP, the role of ISP variability has also been explored. [6] analyzed how varying ISP parameters affect object detection performance and proposed robust data augmentation techniques to enhance model robustness. On the hardware side, [17] developed deep learning accelerators embedded within sensors, achieving higher accuracy but at the expense of increased energy consumption and response time.

B. Transfer Learning for Specialized Applications

Transfer learning has emerged as a powerful technique to address challenges in specialized object detection scenarios, particularly under non-ideal conditions. [18] introduced a feature corrective layer for transfer learning, enabling effective detection of adverse images, such as rainy, dark, or raw images, with marked performance improvements.

In specific domains, [19] applied transfer learning to precision agriculture using lightweight sensors on UAVs, leveraging VisDrone datasets for high efficiency. Similarly, [20] proposed GAIA, a system that efficiently generates customized transfer learning solutions, further demonstrating the adaptability of this approach.

For real-time applications, [21] enhanced the YOLO algorithm with transfer learning to improve its effectiveness in real-world object detection scenarios. Small object detection has also benefited from transfer learning. [22] developed TranSDet, which uses resolution-enhanced training images to improve the adaptability of the model to small objects, while [23] optimized YOLOv3 with pruning and transfer learning to enhance detection performance for small targets.

III. Methodology

To explore the potential of directly utilizing raw Bayer images for object detection, we propose an enhanced approach based on transfer learning, refining the traditional Faster R-CNN framework [11]. The traditional Faster R-CNN architecture (illustrated in Fig.1 (a)) extracts features using a ResNet-50 [24] backbone with a Feature Pyramid Network (FPN) [12],

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and employs a Region Proposal Network (RPN), ROI Pooling, classification, and bounding box regression modules for detection. However, this design relies on RGB images and is not optimized for the unique feature distribution of raw Bayer images. To address this, we introduced modifications to the Faster R-CNN architecture (illustrated in Fig.1 (b)) and designed a preprocessing workflow tailored for Bayer images.

A. Model Structure and Improvement

1) feature extraction and transfer learning: In the modified architecture, we retain the feature extraction backbone (ResNet-50 with FPN) of the traditional Faster R-CNN as the base for transfer learning. To adapt to the distinct feature distribution of Bayer images, we apply a partial freezing strategy: the earlier layers of the backbone remain frozen to preserve general visual features, while the last five layers are fine-tuned to accommodate the specific characteristics of the Bayer image dataset. This strategy allows the model to leverage pre-trained weights while effectively adapting to the new domain, ensuring that high-level features are optimized for raw Bayer images.

2) Region Proposal Network(RPN): The RPN module generates candidate proposals, a critical component of the Faster R-CNN architecture. Given the high noise and dynamic range of Bayer images, we revised the anchor configurations in the RPN, including anchor ratios and scales, to better suit the multi-scale nature of the objects in Bayer images. This ensures robust proposal generation across varying object sizes.

3) ROI Pooling and detector head: The ROI Pooling module, which extracts features for candidate regions, remains unchanged. However, the classification and bounding box regression heads, closely tied to the input features, are fine-tuned through transfer learning. By retraining these heads with features extracted from Bayer images, the modified architecture achieves improved detection performance tailored to the new input domain.

B. Data Preprocessing and Transfer Learning Processes

Before feeding raw Bayer images into the model, we designed a specialized preprocessing workflow:

1) Channel Separation and Normalization: The original Bayer image is combination of four channels of RGGB. We separated it and input into independent channel. At the same time, normalized processing on each channel. To ensure the stability of the feature distribution and preserve the original information.

2) Data Enhance: To address the inherent variability in Bayer images under different lighting conditions, we applied data augmentation techniques, including random cropping, brightness adjustment, and horizontal flipping. These augmentations enhance the model's generalization capability.

3) Transfer Learning Process: The transfer learning process is divided into two stages:

- **Initial Training:** The traditional Faster R-CNN model is pretrained on RGB images to learn foundational visual features.
- **Fine-Tuning:** The pretrained model is transferred to the Bayer image dataset, with fine-tuning applied to the upper layers of the backbone, the RPN, and the detection heads to adapt the model to the new feature distribution.

IV. Experiments

A. Datasets

The KITTI dataset [25] is a widely used benchmark for autonomous driving research, providing diverse image sequences captured in real-world driving environments. The dataset includes high-resolution images with various perspectives and scenarios, such as urban streets, highways, and rural areas. Its comprehensive annotations make it an excellent choice for evaluating object detection methods.

Images in the KITTI dataset have different sizes, the width varies from 1224 to 1242 pixels, and the height varies from 370 to 376 pixels [25]. For simplicity and to facilitate the conversion to Bayer data, we resized all images to a uniform resolution of 300 pixels in width and 90 pixels in height. This resizing ensures computational efficiency while retaining sufficient image detail for object detection tasks.

In order to generate Raw KITTI dataset, each image is transformed into a Bayer pattern representation by dividing it into a grid of 2×2 cells. Each cell retains only one color channel based on its position:

- The **top-left** pixel retains the R channel values, and sets the G and B channel values to 0;
- The **top-right** and **bottom-left** pixels retain the G channel values, and set the R and B channel values to 0;
- The **bottom-right** pixel retains the B channel values, and sets the R and G channel values to 0.

This can be mathematically expressed as:

$$I_{raw}(x,y) = \begin{cases} I^R_{RGB}(x,y), & if(x\%2=0) \cap (y\%2=0) \\ I^R_{RGB}(x,y), & if[(x\%2=1) \cap (y\%2=0)] \\ & \cup[(x\%2=0) \cap (y\%2=1)] \\ I^R_{RGB}(x,y), & if(x\%2=1) \cap (y\%2=1) \\ 0, & otherwise. \end{cases} \tag{1}$$

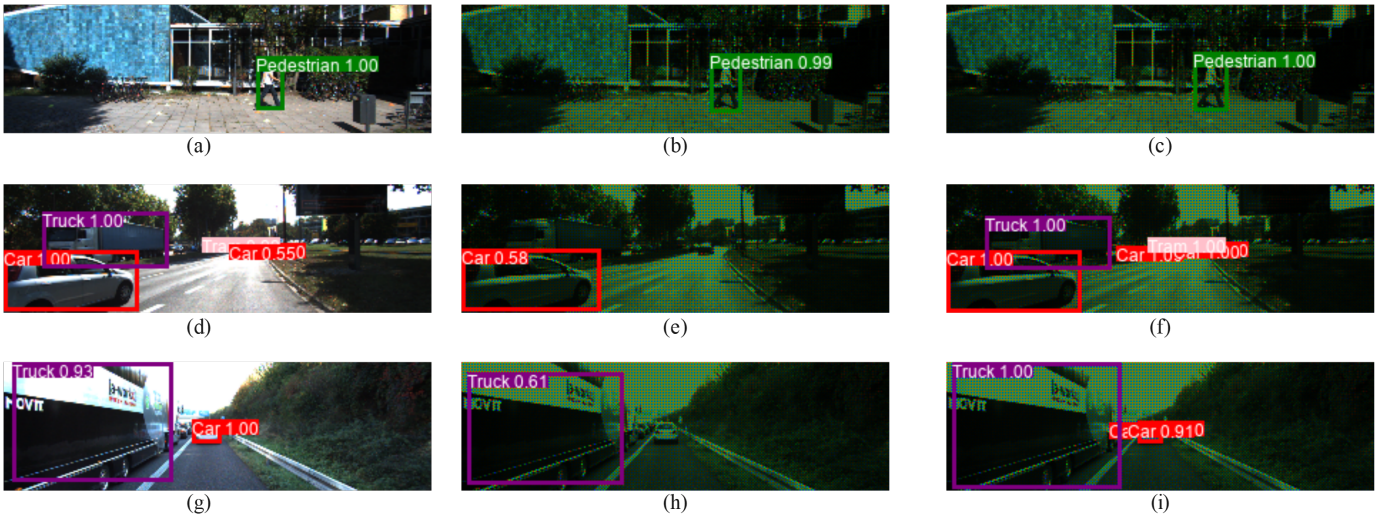
Among these, $I_{raw}(x,y)$ represents the raw Bayer image, and x and y represent the coordinates of the pixels.

I^R_{RGB}, I^G_{RGB}, and I^B_{RGB} correspond to the pixel values of the red, green, and blue channels in the original RGB image I_{RGB}.

The $\%$ operator represents the modulo operation used to determine the position of pixels in a 2×2 grid.

This method effectively converts RGB images to three channel Bayer format, and retain the original image spatial structure. At the same time, simulated the original image output of the camera sensor.

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(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 2. Qualitative Comparison of Object Detection Results on KITTI Dataset. (Detection results on three KITTI images: (a,b,c) for 000000, (d,e,f) for 000016, and (g,h,i) for 000037. Columns represent methods: Faster R-CNN (RGB Data), Faster R-CNN (Raw Data), and Proposed Method (Raw Data).)

TABLE I
COMPARISON OF DETECTION PERFORMANCE ON KITTI AND GENERATED RAW KITTI DATASET

Model	Dataset	mAP		
		@0.5	@0.75	@[0.5:0.95]
Faster R-CNN	KITTI	0.7486	0.4958	0.4568
Faster R-CNN	Generated Raw KITTI	0.7056	0.4238	0.4065
Proposed Method	Generated Raw KITTI	0.7528	0.4656	0.4385

B. Quantitative Results

To validate the effectiveness of our proposed method, we conducted experiments on the KITTI dataset and its generated Bayer dataset (Generated Raw KITTI). Each dataset was randomly divided into 85% for training and 15% for validation, with models trained for 20 epochs under identical settings to ensure fair performance evaluation.

The results, shown in Table I, compare the performance of traditional Faster R-CNN on raw RGB images and generated Bayer images, as well as our method on the generated Bayer images. Our approach builds upon a pre-trained Faster R-CNN model on RGB images, using the structure in Fig. 1. Through transfer learning, we fine-tuned the model with an additional 10 rounds of training on Bayer images by freezing partial parameters and structures.

The results demonstrate that our method significantly outperforms traditional Faster R-CNN on the generated Bayer dataset, highlighting its ability to better adapt to the unique characteristics of Bayer images. Furthermore, while RGB images naturally achieve superior performance, our transfer learning strategy enables Raw images to achieve comparable detection performance, offering an effective solution for processing Bayer images directly.

C. Qualitative Results

The detection results provide an intuitive reflection of model performance. As shown in Fig. 2, we present detection outcomes for three images from the KITTI dataset: 000000, 000016, and 000037. The figure displays results from three methods: Faster R-CNN on RGB images, Faster R-CNN on raw Bayer images, and our proposed method on raw Bayer images.

From the detection results in Figure 2, it can be observed that the performance of Faster R-CNN on RGB images and our proposed method on raw Bayer images are quite similar, which aligns closely with the results reported in Table I. However, Faster R-CNN has a relatively poor detection effect on the raw Bayer image, especially the effect is not ideal in detecting long-distance objects. The decline of detection quality of small target and distant target is obviously, highlighting the advantage of our transfer learning-based approach in adapting to the raw Bayer image domain.

V. CONCLUSION

In this study, we proposed an enhanced approach to adapt the Faster R-CNN architecture for object detection directly on raw Bayer images. By leveraging transfer learning, we fine-tuned a pre-trained Faster R-CNN model to accommodate the distinct characteristics of Bayer images without altering the original architecture. Our results demonstrate that models trained on raw Bayer images can achieve comparable performance to those trained on RGB images, especially in scenarios where ISP-induced artifacts may hinder detection accuracy. Moreover, our method significantly outperforms traditional

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approaches on Bayer images by effectively capturing their unique characteristics.

This study highlights that, with an appropriate training strategy, raw Bayer images can replace RGB images as inputs for object detection tasks without increasing model size or complexity. By bypassing computationally intensive ISP pipelines, this approach offers a promising solution for low-power, resource-constrained edge computing applications. Future work will focus on further optimizing the training process and extending the method to other imaging modalities and real-world scenarios, particularly under adverse conditions.

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